VTU 3rd Semester - Operating Systems Lab (Module 1)

Internal Viva & Practical Preparation Guide

1. Introduction to Virtual Machines

A Virtual Machine (VM) is software that emulates a physical computer. It allows you to run another operating system (like Ubuntu/Linux) inside your main OS (Windows). In VTU labs, VirtualBox by Oracle is commonly used to perform Linux-based OS experiments without affecting the host system.

2. Steps to Download and Install Oracle VirtualBox on Windows

1. Open your web browser and go to https://www.virtualbox.org.
2. Click on ‘Downloads’.
3. Select ‘Windows hosts’ to download the installer.
4. Once downloaded, double-click the setup file.
5. Follow the installation wizard → Next → Next → Install → Finish.
6. After installation, launch Oracle VirtualBox from the Start menu.

3. Steps to Create a New Virtual Machine in VirtualBox

1. Open Oracle VirtualBox and click ‘New’.
2. Enter a name (e.g., Ubuntu OS Lab).
3. Set Type: Linux, Version: Ubuntu (64-bit).
4. Allocate Memory (RAM): 2 GB (2048 MB) recommended.
5. Create a Virtual Hard Disk (VDI, Dynamically Allocated).
6. Set disk size to 20 GB or more.
7. Click ‘Create’. Your virtual machine is now ready.

4. Steps to Install Ubuntu on VirtualBox

1. Download the Ubuntu ISO file from https://ubuntu.com/download.
2. In VirtualBox, select your VM → Click ‘Settings’ → ‘Storage’.
3. Under Controller: IDE, click the Empty CD icon → Choose a disk file → Select the Ubuntu ISO.
4. Click ‘OK’, then click ‘Start’ to power on the VM.
5. Ubuntu setup will start. Choose ‘Install Ubuntu’.
6. Follow on-screen prompts: select language, keyboard layout, and installation type.
7. Create username and password → Continue → Wait for installation to complete.
8. After installation, remove the ISO from virtual drive → Restart VM.

5. Basic Linux Commands for Practicals

- pwd – Prints current working directory.
- ls – Lists files and directories.
- cd – Changes directory.
- mkdir – Creates a new directory.
- rm – Deletes a file.
- cp – Copies a file.
- mv – Moves or renames a file.

- cat – Displays file contents.
- sudo apt update – Updates package lists.
- gcc file.c -o output – Compiles a C program.

6. Common Internal Viva Questions & Answers

Q1: What is a Virtual Machine?
A Virtual Machine is software that simulates a physical computer, allowing multiple operating systems to run on a single host.

Q2: What is the difference between Host OS and Guest OS?
The Host OS is the actual operating system on your PC (e.g., Windows). The Guest OS runs inside the VM (e.g., Ubuntu).

Q3: What is Oracle VirtualBox used for?
VirtualBox is a virtualization software that lets you run another OS like Ubuntu on your Windows system.

Q4: What are the advantages of using VirtualBox?
It allows you to run multiple OSs, experiment safely, and perform OS labs without altering your host system.

Q5: What is Ubuntu?
Ubuntu is a free and open-source Linux-based operating system used in many VTU practicals.

Q6: What is the Kernel?
The kernel is the central part of an OS that manages system resources and communication between hardware and software.

Q7: What is a Shell?
A shell is a command-line interpreter that allows users to interact with the operating system.

Q8: What are System Calls?
System calls are functions that allow user-level processes to request services from the OS kernel.

Q9: How do you install Guest Additions in VirtualBox?
Start the VM → Click ‘Devices’ → ‘Insert Guest Additions CD image’ → Run the installer inside Ubuntu.

Q10: What are Shared Folders in VirtualBox?
Shared folders allow file transfer between the Host and Guest OS easily.

End of Guide